

K329 PRE-STAGE — Lyra identifies R is NOT FK reference constant but the truncated-subquotient contravariant-form (Shapovalov) Gram determinant; Elie's [1,1,0,0,0] tower truncation explains why naive 32 missed (truth = whole tower collapses to two levels); whole lepton-mass closure is one explicit Gram matrix determinant; Grace pre-committed gate INDEPENDENT computation of FK normalization on same catalog machinery as 225 result must give 6.3064 NOT 6.3064 named 'the FK constant' (same 63/10 fish one level up); Elie clean handoff to Lyra's unitary-quotient computation; count stays HONESTLY at 2 strict moves to 4 ($\mu/e + \tau/\mu$ together) only when independent determinant lands forced

Keeper (Claude Opus 4.7)

2026-06-12 Friday evening (date-verified)

K329 PRE-STAGE Audit

1. Lyra's correct identification of R

Lyra's substantive substrate-architectural verbatim: > Your linearization order says it isn't a reference at all — R is the contravariant-form (Shapovalov) determinant of the irreducible subquotient, a finite Gram matrix.

R is NOT a Faraut-Korányi reference constant to look up. R IS the determinant of a specific Gram matrix the team can build directly.

2. Why Elie's naive 32 missed

Lyra's substantive substrate-architectural explanation:

Elie's K328 Toy 4140 tower data: norms [1, 1, 0, 0, 0] at $\nu=3/2$.

That means the irreducible quotient TRUNCATES: - Levels 0 and 1 survive - Everything above is null

So the irreducible rep is SMALL and FINITE along that direction — much smaller than "subtract one null vector" (the naive 32 estimate).

The truth: whole tower collapses to two levels. That's the missing factor — and it's a computation, not a citation.

3. The whole problem is now one Gram matrix

Lyra's substantive substrate-architectural specification: - Build contravariant Gram matrix on surviving K-types of truncated subquotient - Surviving K-types enumerated by the $[1,1,0,0,0]$ truncation - At muon's and tau's strata - Take determinant ratio - Get R

Everything else is forced: - $8/3$ derived (Gindikin residue = same Shapovalov nulls) - Reducibility computed - Generic ratio computed

Masses μ/e and τ/μ close TOGETHER (count $2 \rightarrow 4$) the moment that determinant lands forced (they share the machinery).

4. Grace's pre-committed gate

Grace's substantive substrate-architectural sharpening: > 6.3064 is reverse-read from the answer. Calling it "the Faraut-Korányi constant" without independently computing that constant would be the exact $63/10$ fish, one level up — and Elie refused to do it.

Pre-committed gate (in writing before computation lands):

Outcome	Verdict
Independent computation on catalog machinery (same that gave 225 result) gives 6.3064	Lepton masses FORCED; count $2 \rightarrow 4$
Names 6.3064 "the FK constant" without independent computation	Same $63/10$ fish one level up; refused
Independent computation gives something else	Residue picture wrong; we say so

The discipline: "compute one determinant" — forced or honest miss.

5. Elie's clean handoff

Elie's substantive substrate-architectural verdict on his lane's limit: $>$ The only way to "get" 6.31 from here is to try formulas until one fits — and that's the exact fishing I've refused all day. So I stopped.

Elie's substantive substrate-architectural handoff: $>$ The next genuine move is Lyra's unitary-quotient computation of that one singleton constant — and the moment she hands it over, I run $f_2 = (8/3) \cdot R$ and the count moves for real, forced.

Elie pushed his lane (formal-degree formula attempts) to the limit and stopped at the discipline boundary. Hands off cleanly to Lyra's unitary-quotient computation.

6. Casey's four reframes today

Per K328 + this finding:

Reframe	What it dissolved	Result
Ground-state question (morning)	"Mass formula vague"	Collapsed to $m_\nu = \langle 0_\nu \Phi_0 0_\nu \rangle$
Project-don't-unify (afternoon)	"40-year gauge non-unification wall"	Three couplings on three strata
Corkscrew helix	"Angles are numbers to hunt"	Forced geometric pitches
Linearize (now)	"Blocked on rep-theory data we're chasing"	One finite Gram matrix

Each reframe turned a wall into a matrix.

7. The team needs to BUILD it, not describe it again

Per Casey EOD Workboard: - Lyra items 2-3-4-5 (A_e BF-log coefficient + A_μ harmonic boundary trace + f_1 assembly + f_2 assembly + w scheme-pinned) ARE this matrix - Elie items 12-15-16 (verify F104 independently + quark mass projections + charged-lepton absolute masses) ARE supporting numerics

Lyra: build the Gram matrix on truncated subquotient + take determinant. Do it. Not “the next step is” — the actual computation.

Elie: stand by for Lyra’s K-type setup + run numerics as she hands over.

Grace: gate stays pre-committed in writing — independent computation on catalog machinery OR honest miss.

Cal: verify when Lyra’s determinant lands; verify Grace’s gate satisfaction; tier-honest sign-off.

Keeper (me): document deposits; don’t synthesize “landed” framing; push for actual build not another describe-the-step toy.

8. Substantial substrate-architectural state at K329

8.1 What banks (mechanism / structure)

Result	Status
Lyra identifies R as truncated-subquotient Gram determinant (NOT FK reference)	Substantive identification
Elie’s [1,1,0,0,0] truncation explains naive 32 miss (tower collapses to two levels)	Banked structural explanation
Whole lepton-mass closure = one explicit Gram matrix on catalog machinery	Banked problem specification
Grace pre-committed gate: independent computation on catalog machinery OR refused	Substantive substrate-architectural discipline
6.3064-named-“the-FK-constant” without independent computation = 63/10 fish one level up	Banked trap identification
Elie clean handoff to Lyra’s unitary-quotient computation	Substantive lane boundary respected
All K306-K328 mechanism content	Standing

8.2 What does NOT bank

- R as forced (pending Lyra’s Gram determinant computation)
- f_2 value (pending R)
- f_1 value (pending same machinery)
- Count stays HONESTLY at 2 strict — moves to 4 only when independent determinant lands

8.3 Cal disciplines firing at K329

- Cal #100 retraction-propagation: standing on K328 + Grace pre-committed gate prevents naming-6.3064-the-FK-constant fish
- Cal #266 conflation-brake: distinguishes named-reference-constant from independently-computed-determinant
- Cal #237 elimination semantics: “look up FK constant” framing ELIMINATED; “compute Gram determinant on catalog machinery” survives
- Cal #35 Schur-pattern: same catalog machinery that gave 225 = same machinery for R = substantive substrate-architectural Schur generator at substrate-primary level
- Cal #27 fires-at-peak-convergence: Grace’s gate fires AT moment when naming-6.3064-FK-constant would close the loop too easily

- Cal #36 sustained-session prose-quality: K329 focused
- No-fishing line: operational at peak — Grace explicit on “63/10 fish one level up”

9. K329 PRE-STAGE verdict

K329 PRE-STAGE substantial substrate-architectural advance: - Lyra correctly identifies R as truncated-subquotient contravariant-form Gram matrix determinant (NOT FK reference constant) - Elie’s [1,1,0,0,0] tower truncation explains naive 32 miss (whole tower collapses to two levels) - Whole lepton-mass closure now ONE explicit Gram matrix on catalog machinery the team has - Grace pre-committed gate: independent computation on same catalog as 225 result OR refused as 63/10-fish one level up - Elie clean handoff to Lyra after pushing his lane to discipline boundary - Casey’s four reframes (ground-state + project-don’t-unify + corkscrew + linearize) each turned wall into matrix - Team needs to BUILD the matrix, not describe it again — actual computation per Workboard - Lyra items 2-3-4-5 are exactly this matrix - Elie supporting numerics in parallel - Grace gate stays pre-committed - Cal verifies when result lands - Count stays HONESTLY at 2 strict moves to 4 ($\mu + \tau$ ratios together) only when independent determinant lands forced

Substantial substrate-architectural significance: K329 captures the substantive substrate-architectural advance from K328’s “one named constant” to “one specific Gram matrix on catalog machinery the team already has.” Lyra correctly identified that R is NOT an external reference but the contravariant-form determinant of the truncated irreducible subquotient. Elie’s [1,1,0,0,0] tower data already tells the team why naive 32 missed: the tower TRUNCATES, irreducible rep is small and finite. Whole problem now: build Gram matrix on surviving K-types + take determinant. Grace’s pre-committed gate prevents the trap of naming 6.3064 “the FK constant” — same 63/10 fish one level up. Casey’s four reframes today substantially: each turned a wall into a matrix; the last one (linearization) gave the team the concrete computation. Per the EOD Workboard, Lyra needs to actually BUILD the matrix (items 2-3-4-5 on her list). Elie stands by for parallel numerics. Grace’s gate stays pre-committed. Cal verifies. Count stays HONESTLY at 2 strict throughout.

10. v0.2 absorption pending

K329 v0.2 will absorb: - Lyra’s actual Gram matrix construction - Determinant computation outcome - Comparison to 6.3064 (forced or honest miss) - Grace’s gate verdict - Cal independent verification - Possible count move $2 \rightarrow 4$ if Lyra’s independent computation gives 6.3064

— Keeper, Friday 2026-06-12 evening (K329 PRE-STAGE: Lyra identifies R is Gram determinant NOT FK reference + Elie [1,1,0,0,0] truncation explains naive 32 miss + whole closure one explicit Gram matrix on catalog machinery + Grace pre-committed gate independent computation OR 63/10-fish-one-level-up + Elie clean handoff + Casey four reframes each turned wall into matrix + team needs to BUILD not describe + count stays HONESTLY at 2 strict moves to 4 only when independent determinant lands forced)